# LESSON PLAN

## HELICOPTER THEORY (11)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 02 |
| **Total No. of Lesson Plans** | 27 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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|---|---|
| **1. Topic** | Pressure altitude, density altitude, geometric, absolute altitude and standard altitude |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Pressure altitude, density altitude, geometric, absolute altitude and standard altitude |
| **5. Specific Objectives** | 11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of altitude definitions using IB diagrams; Safety: Ensure no unauthorized access to flight simulators; Supervision: Instructor monitors trainee note-taking and diagram interpretation. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the standard sea-level pressure value?
(b) Define the boundary layer in fluid dynamics.
(c) How does atmospheric pressure change with increasing altitude?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Geometric altitude definition<br>• Absolute altitude calculation<br>• Pressure altitude concept<br>• Standard altitude criteria<br>• Density altitude derivation | L & Ds | PPT, IB | 15 min |
| (b) | • Relationship between pressure and density<br>• Impact of temperature on density altitude<br>• Flight performance implications<br>• Standard atmosphere model<br>• Altitude error sources | L & Ds | PPT, PDF | 20 min |
| (c) | • Comparison of altitude types<br>• Practical application scenarios<br>• Regulatory altitude limits<br>• Pitot-static system basics<br>• Summary of key definitions | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between geometric and absolute altitude.
(b) How does high temperature affect density altitude and aircraft performance?
(c) Define standard altitude and its significance in flight planning.

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| **12. Home Assignment and Follow Up** | Prepare a brief note comparing indicated, pressure, and density altitude, referencing the next lesson on airspeed principles. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Static, dynamic and total pressure, principle of A.S.I, Indicated, rectified, equivalent and true air speed, errors in A.S.I. | Refer to the master syllabus reference list | LP 03 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min